

ECON 1001 — Principles of Microeconomics
Lecture 2 Companion Note
The Amanda Problem: Cost–Benefit Analysis and Opportunity Cost

2026-01-14

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In Lecture 2, I introduced **the Amanda problem**, a challenging example that requires applying the cost–benefit principle and carefully accounting for costs and benefits.

This note reproduces the problem and then breaks the reasoning into **numbered claims**, so you can see how the conclusion follows step by step—and identify exactly where your own reasoning may have gone off track.

Original problem

- 3) Amanda’s favorite band was coming to town. Amanda knew the line to buy a \$20 ticket would take two hours and the value of her time waiting in line is \$25/h. Unfortunately, when she got to the front of the line, the \$20 tickets were sold out (none were left). However, there was a \$50 ticket for a seat in a better location (Amanda preferred that location). Amanda, who also acts rationally, would
- a) Not buy the \$50 ticket.
 - b) Buy the \$50 ticket.
 - c) Call the \$20 ticket a sunk cost.
 - d) [We can’t say what she’d do because we don’t know enough about Amanda’s WTP for the better seat.]

Amanda is at the front of the line, and we are wondering: Will Amanda buy the \$50 ticket?

1. By **the cost-benefit principle**, she will if

$$B_{\$50 \text{ ticket}} > C_{\$50 \text{ ticket}}$$

What is $B_{\$50 \text{ ticket}}$?

It is the enjoyment she gets from “consuming” that ticket—that is, from buying the ticket and seeing the show from that nicer seat.

We are told she prefers the location of the \$50 ticket to that of the \$20 ticket.

2. So $B_{\$50 \text{ ticket}} > B_{\$20 \text{ ticket}}$

Do we know any more? We do know that she chose to buy the \$20 ticket even when it cost her \$20 plus two hours of standing in line, which she values at \$25 each hour (her opportunity cost of time).

This is sometimes called **revealed preference**: by making that choice, she revealed information about how much she valued the ticket.

3. So we know $B_{\$20 \text{ ticket}} > 70$

Combining point 3 with point 2, we have

4. $B_{\$50 \text{ ticket}} > 70$

What is $C_{\$50 \text{ ticket}}$?

Since Amanda is already at the front of the line, there is no additional waiting cost associated with the \$50 ticket. The cost of buying the \$50 ticket is what she gives up to get it. At that point, the cost is simply \$50.

(The time waiting in line is **sunk**: it is nonrecoverable and thus irrelevant.)

5. $C_{\$50 \text{ ticket}} = 50 \$$

Bringing it Together

Combining points 4 and 5 gives

6. $B_{\$50 \text{ ticket}} > C_{\$50 \text{ ticket}} \$$

Which means, she will buy the \$50 ticket.